

Contact.java

```
1 public class Contact {
2     private String name;
3     private String phone;
4
5     //Write a constructor that takes both fields.
6     //Write getters: getName() and getPhone().
7     public Contact(String name, String phone) {
8         this.name = name;
9         this.phone = phone;
10    }
11
12    public String getName() {
13        return name;
14    }
15
16    public String getPhone() {
17        return phone;
18    }
19
20    @Override
21    public String toString() {
22        return name + " | " + phone;
23    }
24
25    public static void main(String[] args) {
26        Contact c1 = new Contact("Bob", "123-456-7890");
27        System.out.println(c1);
28        System.out.println("Name: " + c1.getName());
29        System.out.println("Phone: " + c1.getPhone());
30    }
31 }
32
```

ContactManager.java

```
1 import java.util.*;
2
3 public class ContactManager {
4
5     public static void main(String[] args) {
6         HashMap<String, Contact> contacts = new HashMap<>();
7         contacts.put("Ada Lovelace", new Contact("Ada Lovelace", "+1 617 555 0101"));
8         contacts.put("Bob", new Contact("Bob", "987-654-3210"));
9         contacts.put("Charlie", new Contact("Charlie", "555-555-5555"));
10        contacts.put("David", new Contact("David", "111-222-3333"));
11        contacts.put("Eve", new Contact("Eve", "444-555-6666"));
12
13        //Use contacts.get("Ada Lovelace") to retrieve the contact.
14        if (contacts.get("Ada Lovelace") == null) {
15            System.out.println("Contact not found.");
16        } else {
17            System.out.println(contacts.get("Ada Lovelace"));
18        }
19
20        if (contacts.get("Subramani") == null) {
21            System.out.println("Contact not found.");
22        } else {
23            System.out.println(contacts.get("Bob"));
24        }
25
26        //Create an ArrayList from the HashMap's values:
27        ArrayList<Contact> sorted = new ArrayList<>(contacts.values());
28        //ArrayList<Contact> sorted = new ArrayList<>(contacts.values());
    }
}
```

```
29 | //Sort it alphabetically by name using this one line:
30 |         sorted.sort((a, b) -> a.getName().compareTo(b.getName(
    | )))
31 | //Loop through the sorted list and print each contact.
32 |         System.out.println("print contact in sorted list below
    | :");
33 |         for (Contact contact : sorted) {
34 |             System.out.println(contact);
35 |         }
36 |     }
37 |
38 |
39 | }
40 |
41 |
```